



Zoom-whirl orbital dynamics and Kludge gravitational waveforms in extreme mass-ratio inspirals near charged black holes surrounded by perfect fluid dark matter

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Mathematics, rightly viewed, possesses not only truth, but supreme beauty—a beauty cold and austere ... yet sublimely pure and capable of stern perfection such as only the greatest art can show.

Bertrand Russell

Prerequisite

1. EMRIS- Extreme Mass Ratio Inspiral System

Mass requirement

$$10^7 M_{\odot} + 10 M_{\odot}$$

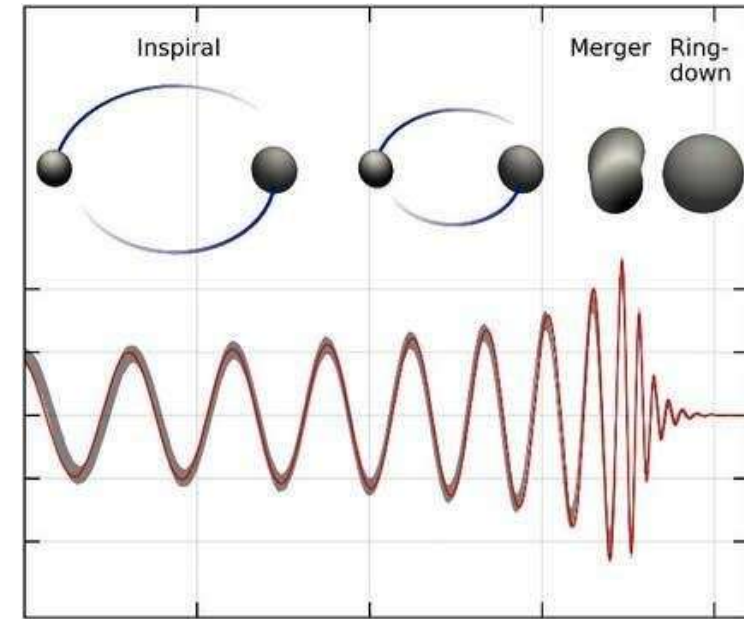
Possible BH systems

1. Supermassive BH- Stellar Mass BH
2. Supermassive BH- Blue Supergiant
3. Supermassive BH- Main Sequence Stars

Test mass approximation

$$\frac{m}{M} \sim 10^{-6}$$

Time-like Geodesic analysis is valid.



Description of possible stages of merger and GW structure [1]

2. Charged BH with PFDM

$$ds^2 = -f(r)dt^2 + \frac{1}{f(r)}dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

$$f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2} + \frac{\gamma}{r} \ln \frac{r}{|\gamma|} \quad [2]$$

1. Al Mubaideen, Hisham. (2023). Theoretical Estimation of the Parameters of the Black Hole - Neutron Star Merger GW200115 via Inspiral, Merger and Ringdown IMR Numerical Relativity Frameworks.
2. A. Das, A. Saha, S. Gangopadhyay, Investigation of circular geodesics in a rotating charged black hole in the presence of perfect fluid dark matter, Classical Quantum Gravity 38 (6) (2021) 065015.

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Spacetime Model

Charged Schwarzschild-type black hole surrounded by Perfect fluid dark matter

Mass

M

Charge

Q

PFDM

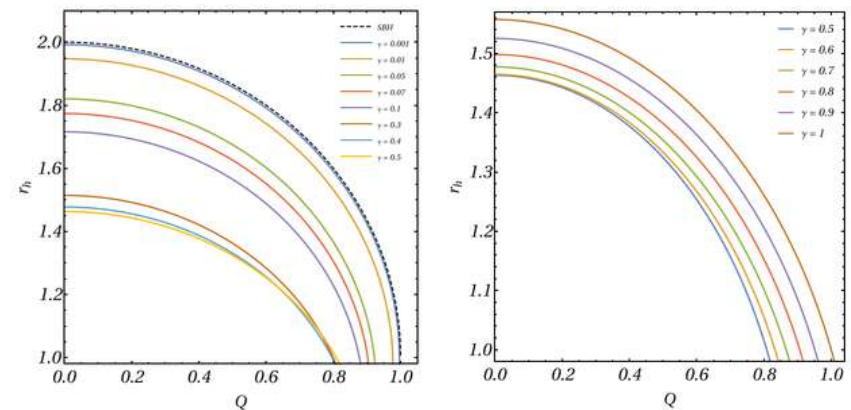
γ

PFDM Parameter: $\gamma = 8\pi r^3 \rho$

Horizon Radius: $r_h^2 - 2Mr_h + Q^2 + \gamma r_h \ln \frac{r_h}{|\gamma|} = 0$

$$r_{hm} = \frac{e}{1+e} \left(M + \sqrt{M^2 - \left(1 + \frac{1}{e}\right) Q^2} \right)$$

$$\gamma_c = \frac{1}{1+e} \left(M + \sqrt{M^2 - \left(1 + \frac{1}{e}\right) Q^2} \right)$$



Horizon Plots

2. Periodic orbit

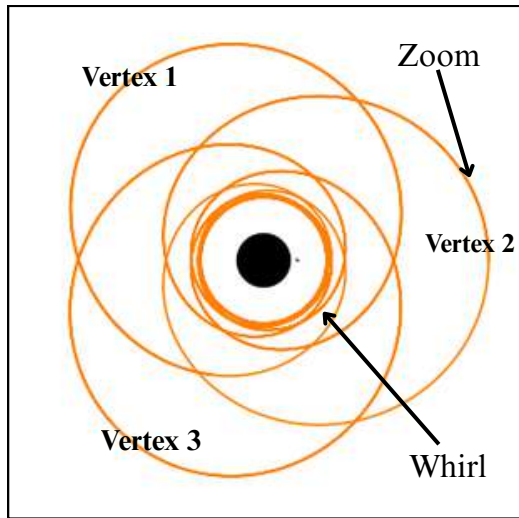
MBO: $V_{eff} = E^2 = 1, \frac{dV_{eff}}{dr} = 0$

ISCO: $V_{eff} = E^2, \frac{dV_{eff}}{dr} = 0, \frac{d^2V_{eff}}{dr^2} = 0$

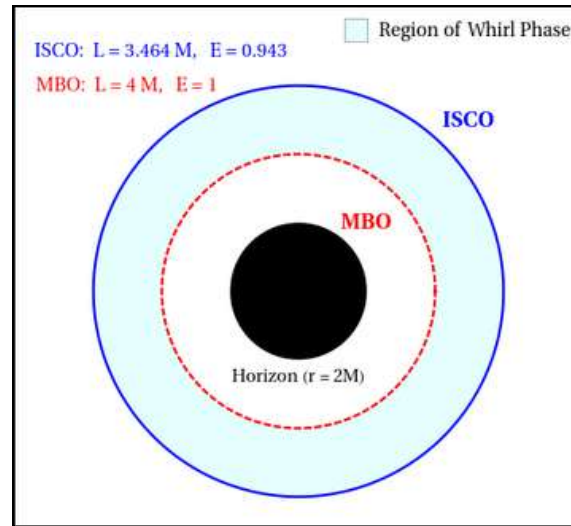
$$E_{ISCO} < E < E_{MBO}$$

$$L_{ISCO} < L < L_{MBO}$$

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Zoom-whirl periodic orbit for particle dynamics around SBH.



Region of whirl phase for zoom-whirl periodic orbit, cyan, for particle dynamics around SBH.

Levin method of zoom-whirl periodic orbit

Taxonomy: (z, w, v)

q value: $q = \frac{\Delta\phi}{2\pi} - 1 = w + \frac{v}{z}, \quad q = \frac{1}{\pi} \int \frac{L}{r^2 \sqrt{E^2 - V_{eff}}} dr - 1$

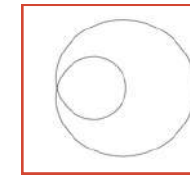
A Periodic Table for Black Hole Orbits

Janna Levin^{*,†} and Gabe Perez-Giz^{**}

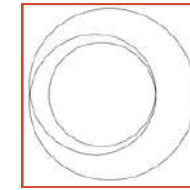
<http://arxiv.org/abs/0802.0459v1>

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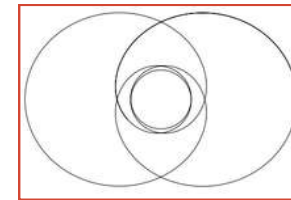
Periodic orbit



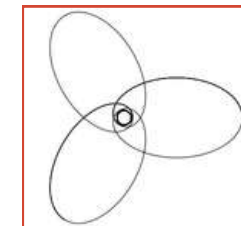
$(1, 1, 0)$



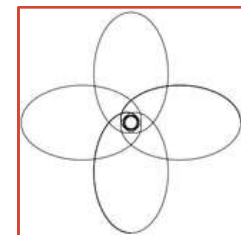
$(1, 2, 0)$



$(2, 1, 1)$



$(3, 1, 2)$

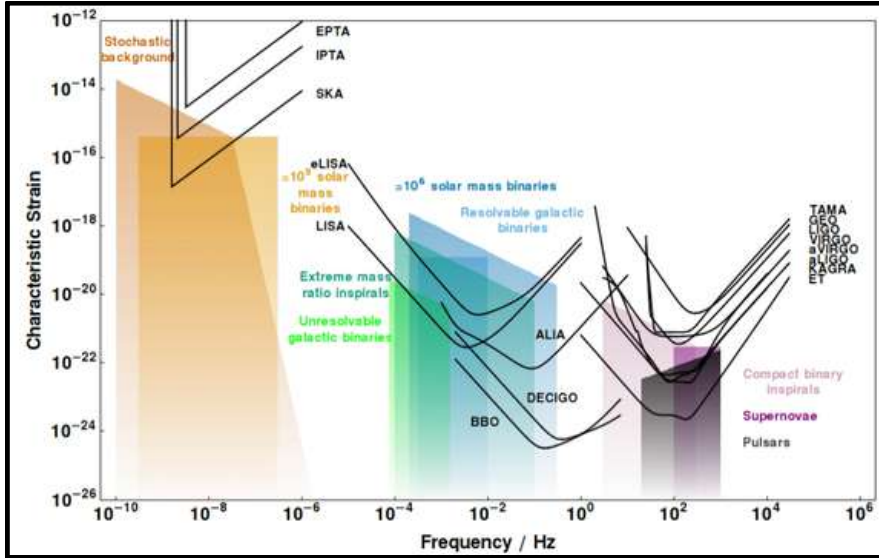


$(4, 1, 3)$

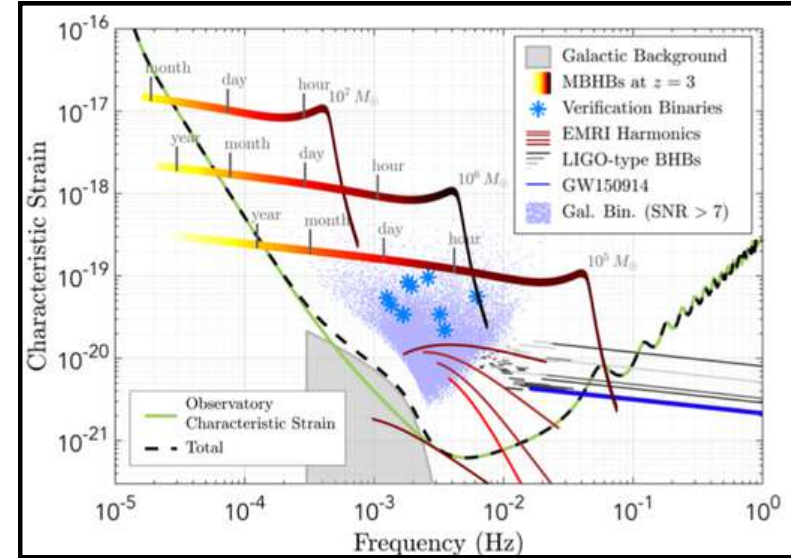
Motivation



Detectability of GW through upcoming space-based observatories such as LISA, Taiji, TianQin, and DECIGO.



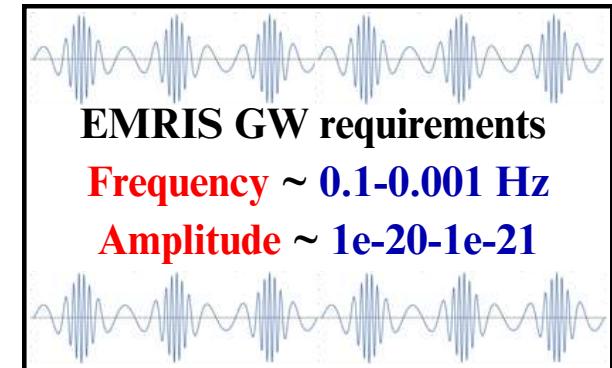
Characteristic strain vs frequency for detectors [1]



Characteristic strain Vs frequency for LISA [2]

Why periodic orbits?

- Systematic method of theoretical analysis
- Long-run observation
- An approximate study of non-periodic orbits and their GW



1. C J Moore et al 2015 Class. Quantum Grav. 32 015014, 10.1088/0264-9381/32/1/015014
2. Travis Robson et al 2019 Class. Quantum Grav. 36 105011, 10.1088/1361-6382/ab1101

Mathematical Modelling*



Lagrangian is given as, $l = \frac{1}{2} g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}$

The value of the conserved quantity comes to be

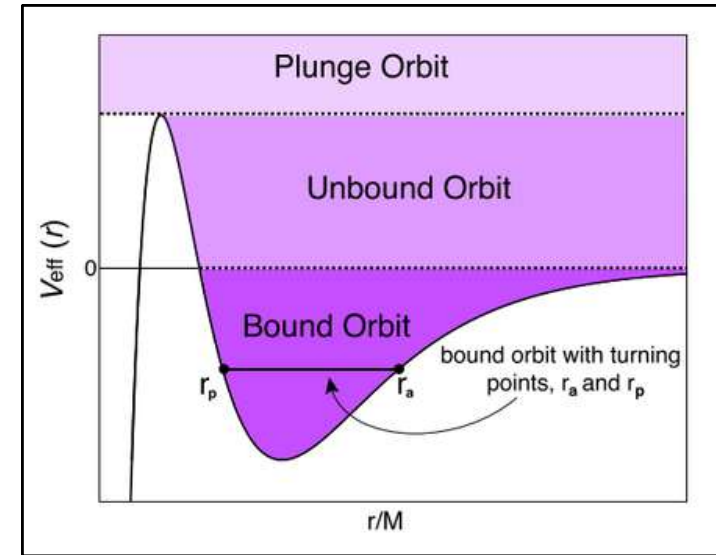
$$E = \frac{dt}{d\tau} f(r) \quad L = -\frac{d\phi}{d\tau} r^2$$

Normalised condition for timelike geodesics

$$g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = -1$$

Now, as we know, that

$$E^2 = V_{eff} + \left(\frac{dr}{d\tau} \right)^2$$



Region for bound orbit for particle dynamics around BH [1].

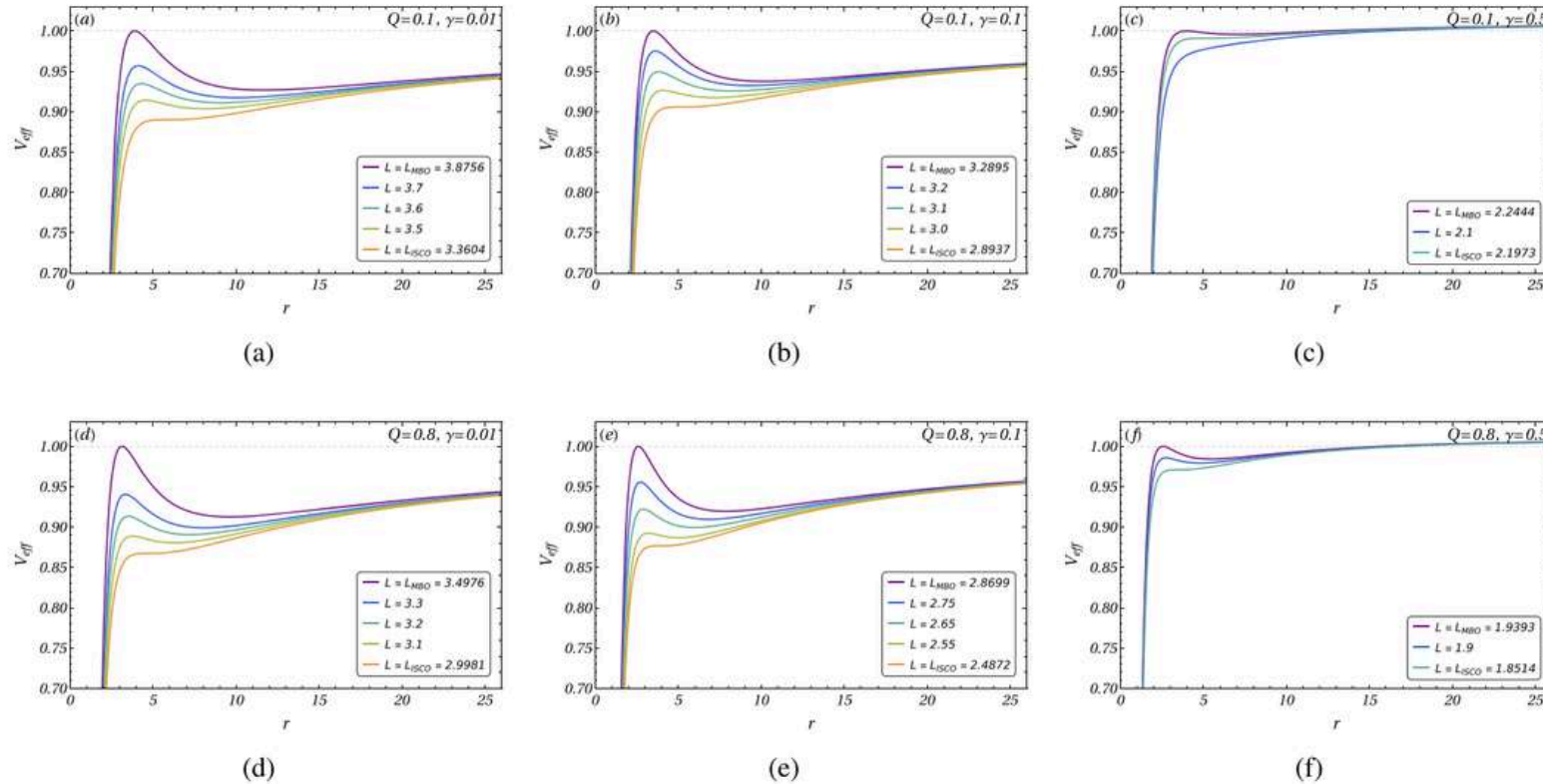
And the value of the effective potential turns out to be

$$V_{eff} = f(r) \left(1 + \frac{L^2}{r^2} \right)$$

*One can find all the required mathematics in chapter 8, 'Geodesics' of Gravity: An Introduction to Einstein's General Relativity, J. B. Hartle (2003).

1. Pakiela, S. (2024). Understanding exotic black hole orbits using effective potentials.

Potential Analysis

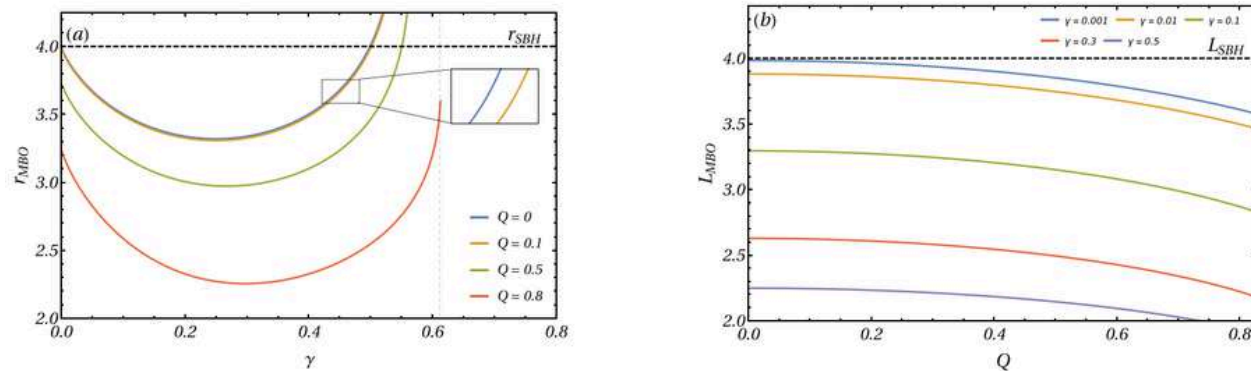


Effective potential plots for orbits between MBO and ISCO

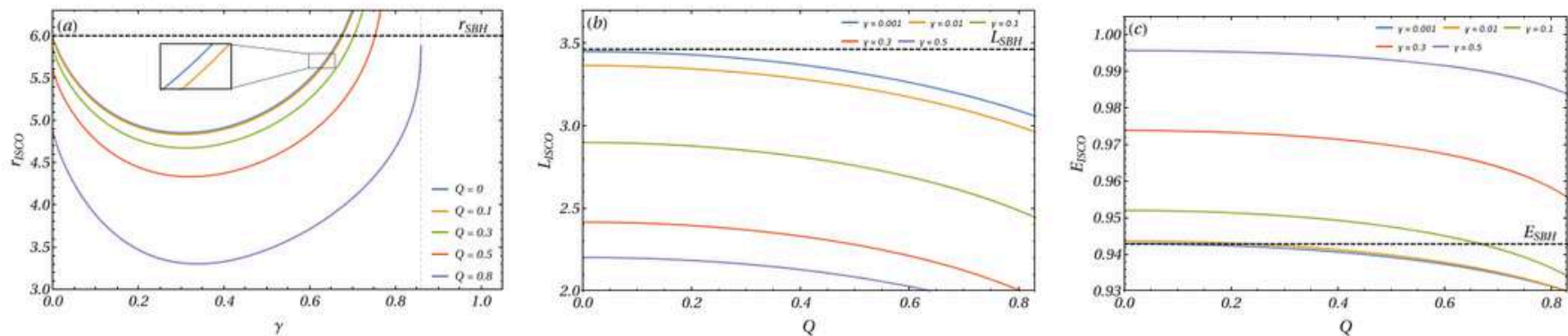
What do charge and PFDM do?

- * More dip in potential with increasing charge, (less energy requirement).
- * Less possible values of energy with PFDM.
- * More possibility of existence of bound orbit for less value of PFDM.

MBO-ISCO Analysis



MBO radius and angular momentum plots

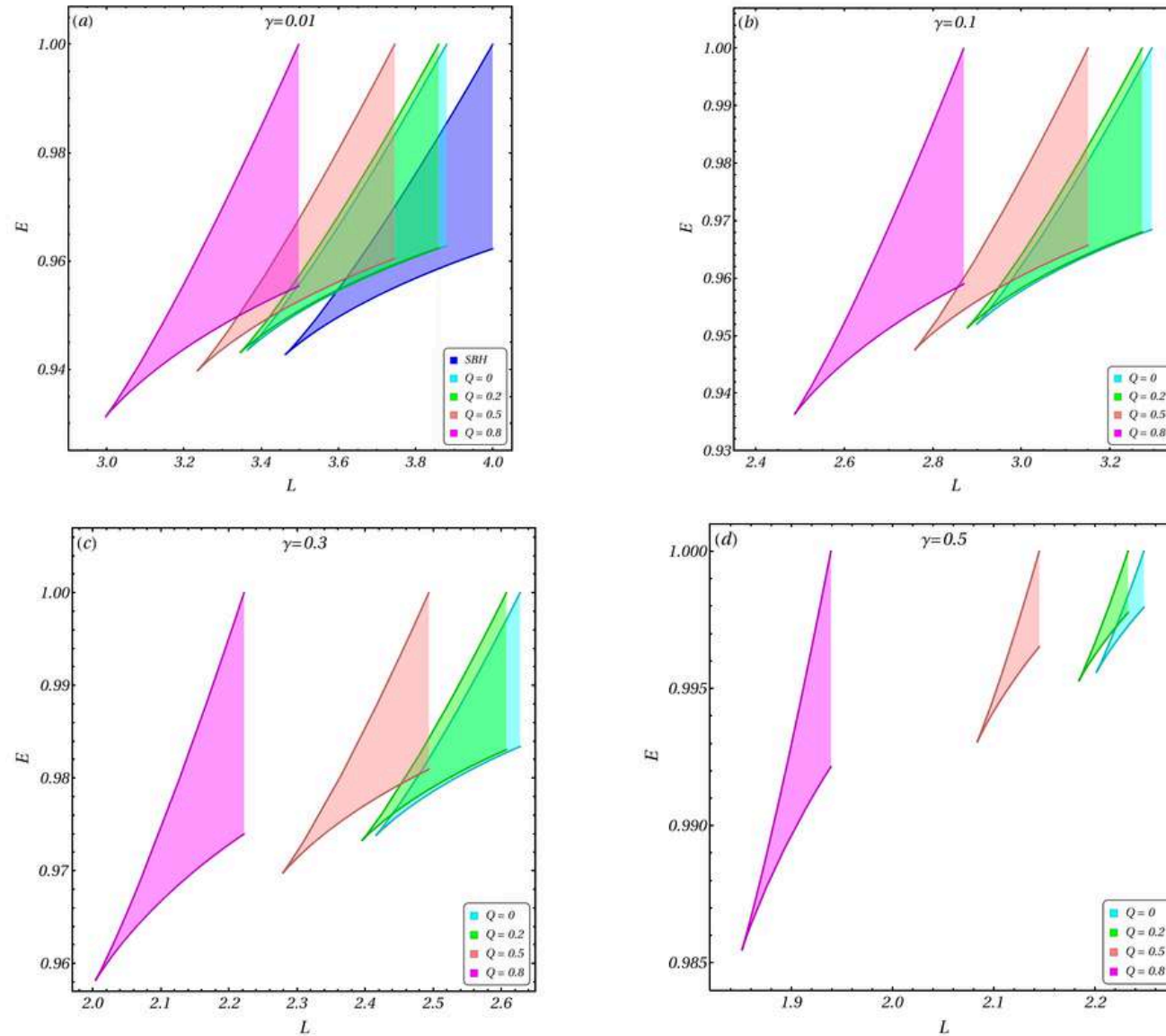


ISCO radius, angular momentum and energy plots

What's unique?

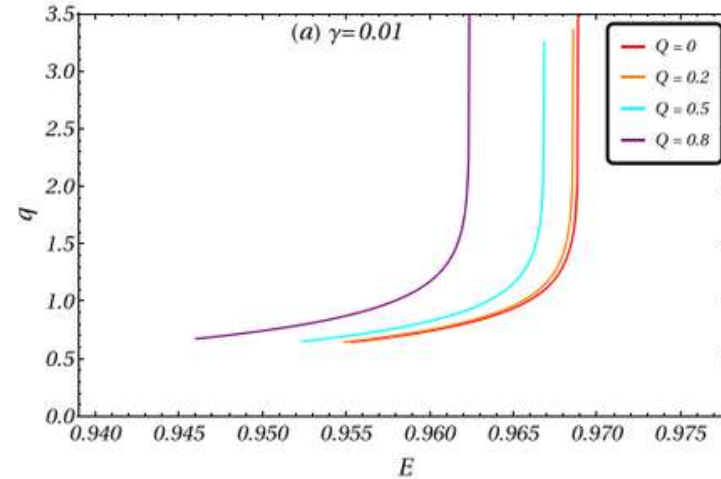
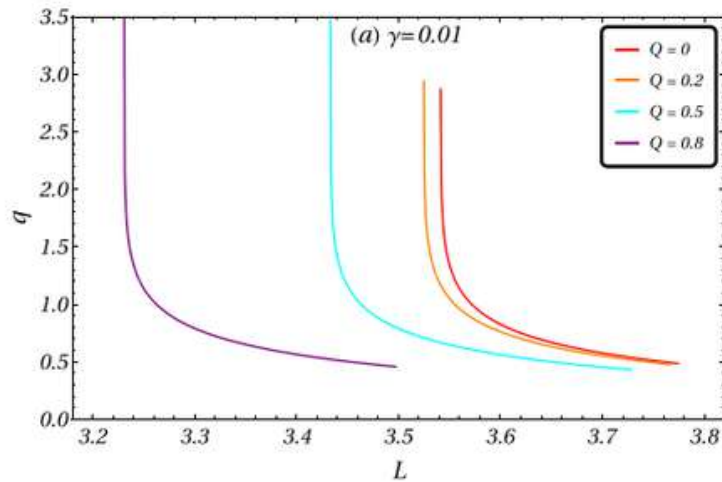
- * Elevating and retarding behaviour of radius with PFDM parameter.
- * Angular momentum decreases with charge and PFDM parameter.
- * ISCO energy decreases with charge and increases with PFDM parameter.

E-L region Analysis



E-L region plot for various values of charge and PFDM parameter.

q-value Analysis



The q-value plot: here (a) q vs L, (b) q vs E. The same plot structure is revealed for other values of the PFDM parameter.

What do we understand?

- * q-value decreases with increase in L.
- * q-value increases with increase in E.
- * Plays an important role for the manifestation of periodic orbit.

Some fun works

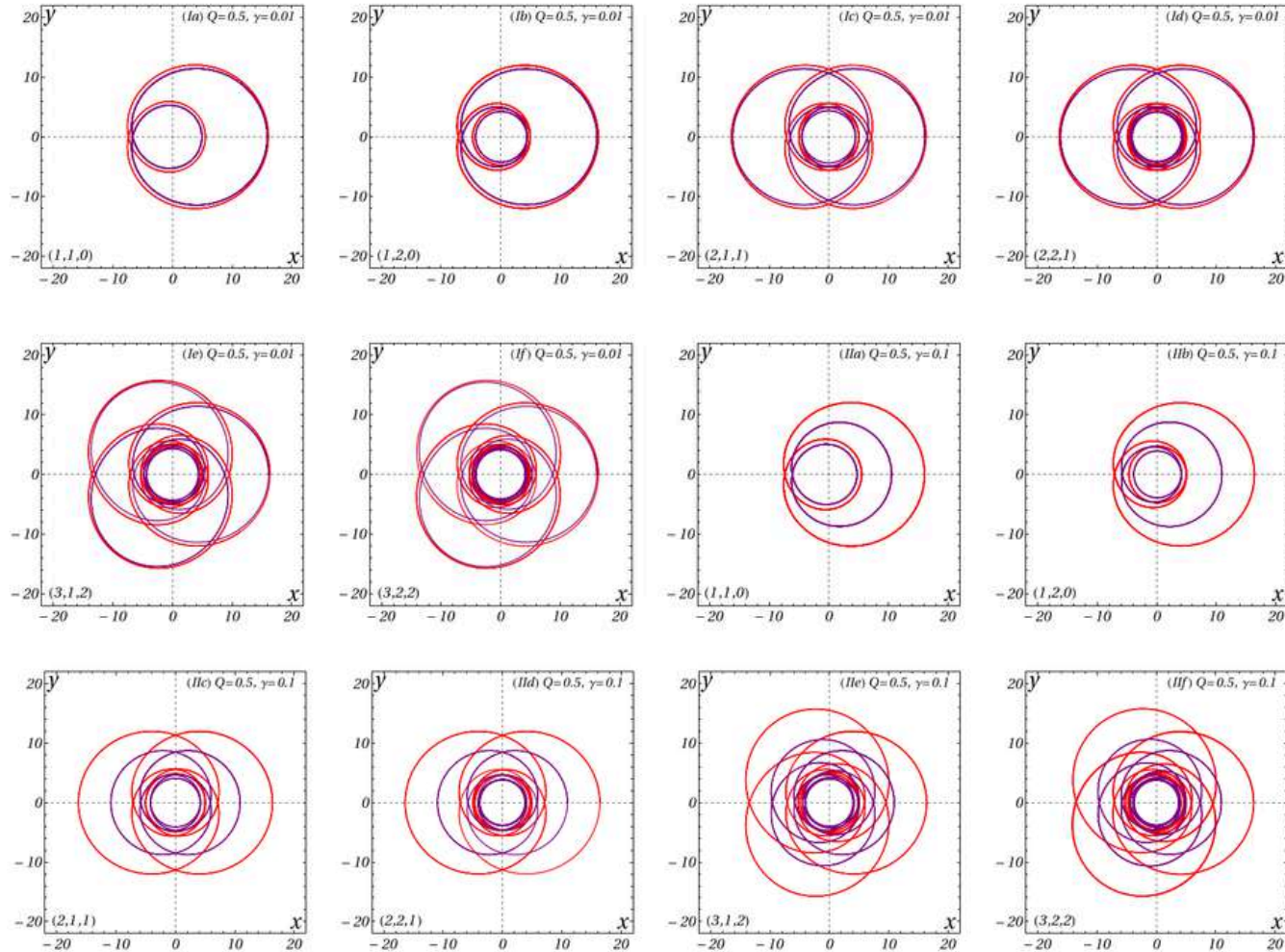
1. Zhao, L., Tang, M. and Xu, Z., 2025. Periodic orbits and gravitational wave radiation in short-hair black hole spacetimes for an extreme mass ratio system. The European Physical Journal C, 85(1), p.36.
2. Bravo-Gaete, M., Lin, J., Liu, Y. and Zhang, X., 2026. Periodic orbits and gravitational waveforms of spinning particles in nonlocal Gravity. arXiv preprint arXiv:2602.15609.
3. Agrawal, R., Kar, A., Jana, S. and Kar, S., 2026. Gravitational wave radiation from periodic orbits in regular black holes. arXiv preprint arXiv:2602.20745.

$$q = w + \frac{v}{z},$$

↓

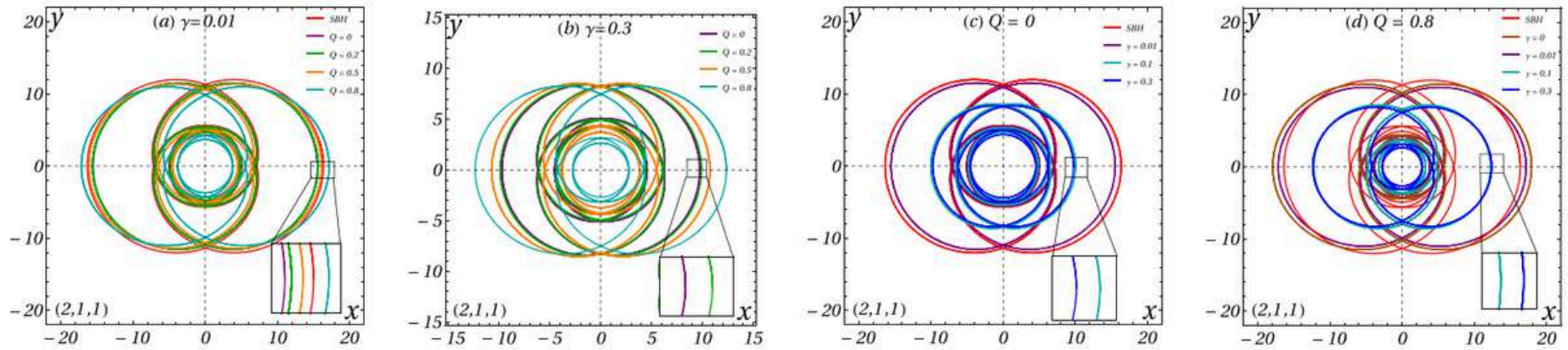
$$q = \frac{1}{\pi} \int \frac{L}{r^2 \sqrt{E^2 - V_{eff}}} dr - 1$$

Periodic Orbits



Periodic orbit plot for various q -values. Here, the purple curve represents periodic motion associated with the SBH spacetime, whereas the red curve represents a periodic orbit associated with a perturbed BH. The energy of the test particle is considered to be $E = 0.96$.

Continue...



The trajectory of a neutral orbiting mass in a charged BH spacetime with PFDM associated with the periodic motion (z, w, v) , between MBO and ISCO for various scenarios, varying Q and constant PFDM, as (a) 0.01, (b) 0.3 and varying PFDM and constant Q as (c) 0, (d) 0.8. The energy value considered for PFDM = 0.01 and 0.1 is $E = 0.96$, and for PFDM = 0.3 is $E = 0.98$.

γ	Q	Distance between the turning points					
		(1, 1, 0)	(1, 2, 0)	(2, 1, 1)	(2, 2, 1)	(3, 1, 2)	(3, 2, 2)
0.01	0.0						
	0.2						
	0.5						
	0.8						
0.1	0.0						
	0.2						
	0.5						
	0.8						
0.3	0.0						
	0.2						
	0.5						
	0.8						

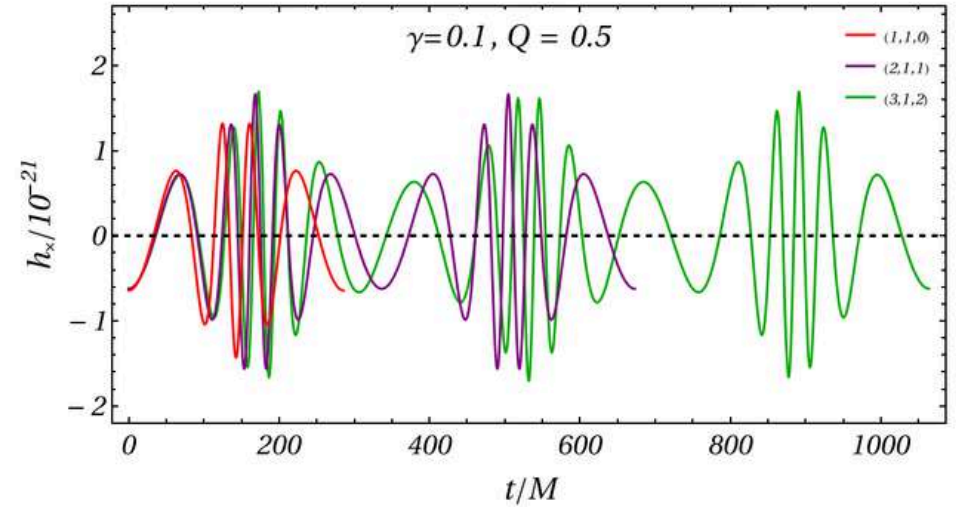
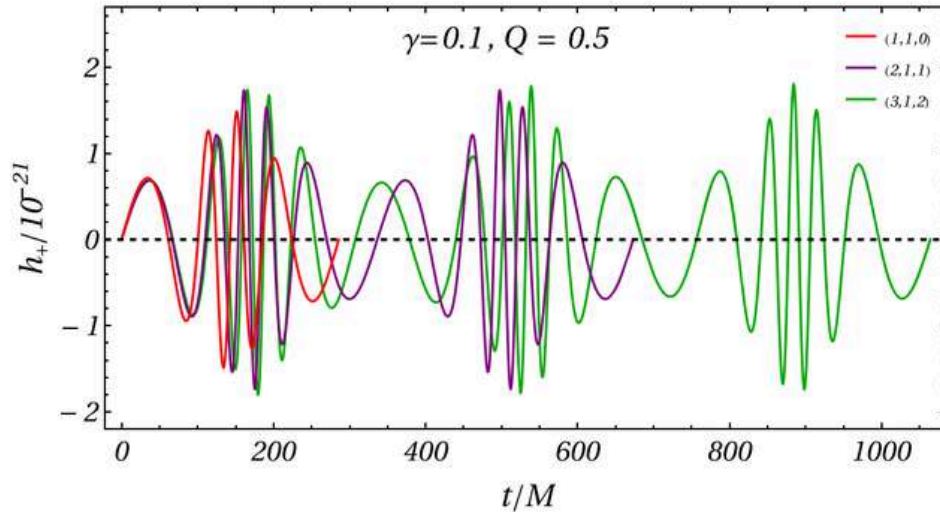
Pictorial representation of the variation of the distance between the turning points.

Gravitational Waves



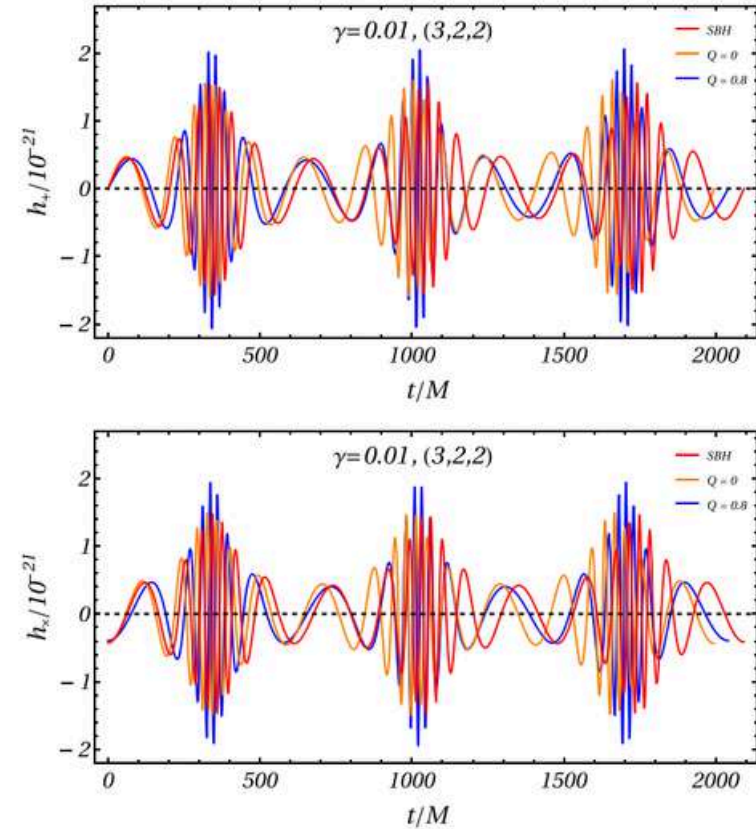
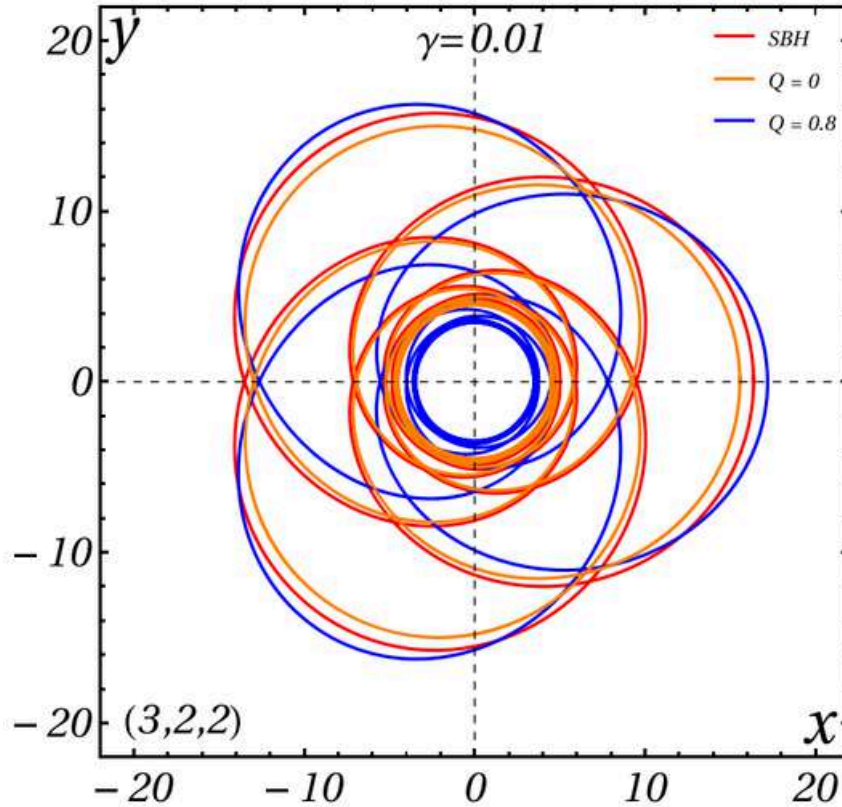
$$h_+ = -\frac{2\mu M^2}{Dr} (1 + \cos^2 \zeta) \cos(2\phi + 2\eta), h_\times = -\frac{4\mu M^2}{Dr} \cos \zeta \sin(2\phi + 2\eta), \mu = \frac{Mm}{(M+m)}$$

For the study of GWs associated with the diverse zoom-whirl orbits, we assume the following parameters for the EMR system: mass of BH, $M = 1e+7 M_\odot$, mass of celestial body, $m = 10 M_\odot$, inclination angle and latitude angle, $\zeta = \eta = \frac{\pi}{4}$ and the luminosity distance, $D = 200$ Mpc.



The gravitational radiation waveforms plotted against dimensionless time t/M for a massive test particle with various orbital trajectories classified by $(1, 1, 0)$, red curve, $(2, 1, 1)$, purple curve and $(3, 1, 2)$, green curve, around a supermassive charged BH suspended in PFDM with values of 0.5 and 0.1. The orbital trajectory is computed by setting $M = 1$, whereas the physical mass values are used to calculate the strain amplitude. The waveforms are represented by the polarisation modes, i.e. h_+ and $h_\times$.

Continue...



The orbital trajectory (left side) and gravitational waveforms (right side) are plotted against dimensionless time t/M for a massive test particle with an orbital trajectory classified by $(3, 2, 2)$ around a supermassive charged BH suspended in PFDM with values of 0.8 and 0.01 . The plot is in comparison to the SBH, red curve, and the SBH with PDFM, i.e., 0 , orange curve. The orbital trajectory is computed by setting $M = 1$, whereas the physical mass values are used to calculate strain amplitude. The waveforms are represented by polarisation modes, i.e. h_+ and $h_{\times}$.

Results



- * Periodic trajectory depends upon the parameters.
- * Elevating and retarding behaviour with PFDM parameter.
- * Study is consistent throughout the analysis.
- * Each orbit has a unique GW structure.
- * GW has high frequency zones associated with whirl and low frequency zones with zooms.
- * GW lie in observable region.

Approximations

- * Adiabatic approximation considered.
- * No high order in the GW.

Discussion



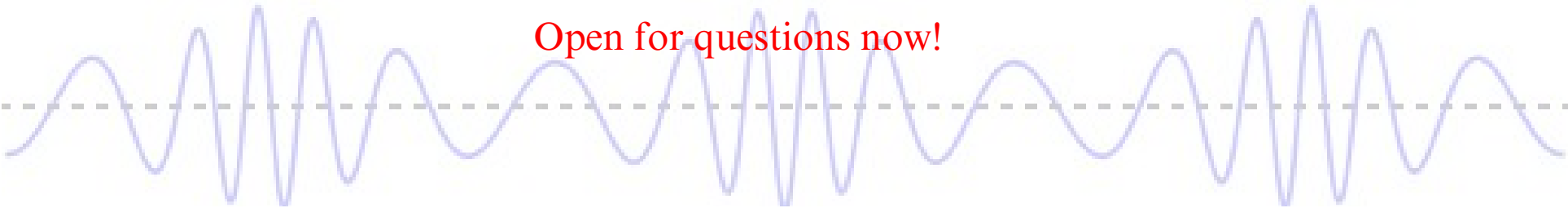
* Can explain physical BH systems?

* Possible future work!



Let's see where analysis like these leads us!

Thank you!!!



Open for questions now!